

View Reviews

Paper ID	360
Paper Title	The Limitations of SMOTE and Class Imbalance in Low-Sample Radiomics: A Cautionary Case Study in Renal Mass Differentiation

Reviewer #3

Questions

1. Overall evaluation. Please provide a detailed review, including a justification for your scores.

In the study, RCC/AML classification is investigated using an SVM-based classifier on 3D radiomics derived from a small and imbalanced CT dataset, with data augmentation via SMOTE. The performance of the method is reported, but no comparison with alternative methods is provided. Moreover, although the title emphasizes the limitations of SMOTE, there is no control condition without SMOTE. Without removing SMOTE and running the same pipeline, it is not possible to disentangle whether the observed instability arises from SMOTE or from the small sample size.

Because the number of positive samples in the dataset is limited, in nested cross-validation the positive class is detected with substantial variability across outer folds in the confusion matrices, leading to high AUC variability. This is an expected outcome given the nature of the dataset.

In Table 1, instead of highlighting only the best fold per split, the mean $\pm$ standard deviation across folds should be reported.

For the other evaluated classifiers (Random Forest, XGBoost, KNN, Logistic Regression), no numerical results are provided; it is only stated that SVM performs best.

In the references, sources [10] and [15] correspond to the same study. In addition, the correct reference should be provided for SMOTE interpolation in the text (citation [15]).

The keyword "radiomics" is listed twice. Additionally, too many keywords are defined.

"decision boundry" $\rightarrow$ "decision boundary"

"Confussion Matrix" $\rightarrow$ "Confusion Matrix"

Reviewer #4

Questions

1. Overall evaluation. Please provide a detailed review, including a justification for your scores.

The paper's strengths lie in its use of a novel dataset that contributes to the field and its status as a valuable study within the domains of medical data processing and artificial intelligence. However, it is recommended that the methods applied to the dataset be illustrated within the text using sample CT images.